1). The highest pH formulation evaluated, Pax-3, provided up to 6-fold more flux than in the absence of NaOH.

Example 23

[0410] An in vitro skin permeation study was conducted using three galanthamine hydrobromide transdermal systems, designated Gala-1, Gala-2 and Gala-3, the compositions of which are set forth in Table 78.

[0411] Round disc samples were prepared in a manner similar to that described in the Methods section, except that the formulation was dried at a temperature of 65° C. and the discs were cut into discs having a diameter of 9/16 inch.

[0412] The theoretical percent weight for each ingredient after drying (calculated assuming all volatile ingredients were completely removed during drying) is set forth in Table 79.

TABLE 78

Component Weight and Weight Percent Based on Total Solution Weight			
	Gala-1 g (wt %)	Gala-2 g (wt %)	Gala-3 g (wt %)
Galanthamine HBr	0.40 (4.7)	0.40 (4.6)	0.40 (4.6)
DI Water	0.30 (3.5)	0.34 (3.9)	0.38 (4.3)
NaOH	0	0.04 (0.5)	0.08 (0.9)
Glycerin	1.00 (11.6)	1.00 (11.5)	1.00 (11.4)
Benzyl Alcohol	0.40 (4.7)	0.40 (4.6)	0.40 (4.6)
PIB adhesive (30% solid)	6.00 (69.8)	6.00 (69.1)	6.00 (68.5)
n-Heptane	0.50 (5.8)	0.50 (5.8)	0.50 (5.7)

TABLE 79

Weight and Theoretical Weight Percent Based on Dried Film Weight			
	Gala-1 g (wt %)	Gala-2 g (wt %)	Gala-3 g (wt %)
Galanthamine HBr	0.40 (11.1)	0.40 (11.0)	0.40 (10.9)
NaOH	0	0.04 (1.1)	0.08 (2.2)
Glycerin	1.00 (27.8)	1.00 (27.5)	1.00 (27.2)
Benzyl Alcohol	0.40 (11.1)	0.40 (11.0)	0.40 (10.9)
PIB adhesive	1.80 (50.0)	1.80 (49.5)	1.80 (48.9)

[0413] Since galanthamine HBr is an acid addition salt of a free base, it reacts with NaOH. The concentration of NaOH in the system after the reaction is completed depends on the amount of galanthamine HBr added. The remaining NaOH concentration after the reaction is completed is defined as the excess NaOH concentration, and was calculated as described in Example 2. The pH was measured as described in the Methods section but using a 2.4 cm^2 circular patch. The pH of the galanthamine HBr patch increased from 8.73 to 10.56 when the calculated excess NaOH concentration in the dried patch was increased from 0% to 1.0%. The pH of the formulation without NaOH was 6.53.

TABLE 80

Excess NaOH Concentration (wt %) and pH			
	Gala-1	Gala-2	Gala-3
Excess NaOH Concentration	—	0%	1.0%
pH	6.53	8.73	10.56

[0414] The in vitro permeation of galanthamine HBr through human cadaver skin from these discs was measured as described in the Methods. Three diffusion cells were used for each formulation. The receiver solution, 5% ethanol/95% PBS buffer (0.05 M KH_2PO_4 with 0.15 M NaCl, pH adjusted to 6.5), was completely withdrawn and replaced with fresh receiver solution at each time point. The samples taken were analyzed by an HPLC for the concentration galanthamine HBr in the receiver solution. The cumulative amount of galanthamine HBr that permeated across the human cadaver skin was calculated using the measured galanthamine HBr concentrations in the receiver solutions.

TABLE 81

Cumulative Amount of Galanthamine HBr (mg/cm^2)			
Time	Gala-1	Gala-2	Gala-3
6 hours	0.140	0.480	0.475
18 hours	0.429	1.058	1.412
24 hours	0.624	1.254	1.750

[0415] The cumulative amount of galanthamine HBr across human cadaver skin at 24 hours increased from 1.254 mg/cm^2 to 1.750 mg/cm^2 when the calculated excess NaOH concentration in the dried patch was increased from 0% to 1.0% as compared to 0.624 mg/cm^2 for the formulation without NaOH.

[0416] The formulation of Gala-2 provided up to 2-fold more galanthamine HBr flux than in the absence of NaOH (Gala-1). The highest pH formulation evaluated, Gala-3, provided up to 3-fold more flux than in the absence of NaOH.

Example 24

[0417] An in vitro skin permeation study was conducted using three hydromorphone hydrochloride transdermal systems, designated Hymo-1, Hymo-2 and Hymo-3, the compositions of which are set forth in Table 82.

[0418] Round disc samples were prepared as described in the Methods section, except that the formulation was dried at a temperature of 65° C. and the discs were cut into discs having a diameter of 9/16 inch.

[0419] The theoretical percent weight for each ingredient after drying (calculated assuming all volatile ingredients were completely removed during drying) is set forth in Table 83.

TABLE 82

Component Weight and Weight Percent Based on Total Solution Weight			
	Hymo-1 g (wt %)	Hymo-2 g (wt %)	Hymo-3 g (wt %)
Hydromorphone HCl	0.20 (2.8)	0.20 (2.7)	0.20 (2.7)
DI Water	0.30 (4.1)	0.38 (5.1)	0.43 (5.7)
NaOH	0	0.08 (1.0)	0.13 (1.7)
Glycerin	1.25 (17.2)	1.25 (16.9)	1.25 (16.7)
PIB adhesive (30% solid)	5.00 (69.0)	5.00 (67.6)	4.00 (66.7)
n-Heptane	0.50 (6.9)	0.50 (6.8)	0.50 (6.7)